

CS 203: Software Tools & Techniques for AI
IIT Gandhinagar
Sem-II - 2024-25

LAB 02

Total: 10 Marks

Submission deadline: Sunday, 19/01/2025 11:59:59 PM

Submission guidelines:

1. Implement the code in google colab and share the link during submission.
2. The deadline will not be extended at any cost.

Note: Submitting this assignment solution confirms that you will follow the IITGN's honor code. We shall strictly penalize the submissions containing plagiarized text/code.

Objective

This assignment aims to familiarize students with modern data validation techniques using [Pandera](#) and [Great Expectations](#) frameworks through hands-on experience with real-world fitness data. Students will learn to implement various validation rules, from basic data type checking to complex cross-column validations, while understanding the strengths and limitations of each framework.

Dataset Preparation

1. Download the data from
“<https://s3.amazonaws.com/tripdata/202412-citibike-tripdata.zip>” using the curl or wget command.
2. Key fields to validate:
 - ride_id (string)
 - rideable_type (categorical)
 - started_at (str)
 - ended_at (str)
 - start_station_name (string)

- start_station_id (string)
- end_station_name (string)
- end_station_id (string)
- start_lat, start_lng (float)
- end_lat, end_lng (float)
- member_casual (categorical)

Features to Implement

1. Pandera Validation Rules:

- Validate the datatype of each column. [20%]
- Analyze each feature and write an appropriate check/scheme for them.
For example, for "rideable_type" check the unique options and ensure that no other entry passes the check. [20 %]
- Add data validation rules to verify that a ride's end time occurs after its start time using Pandera's decorator functionality. [10 %]

2. Great Expectations Suite:

- Validate the datatype of each column by creating an expectation suite. [20 %]
- Implement an Action that will send a mail when encountering a failure.
Take a screenshot of the mail and attach it to the submission file. [20 %]

3. Code Quality:

- Maintain **clean, modular**, and readable code. [5 %]
- Provide comments where necessary. [5 %]

*****Best of Luck*****